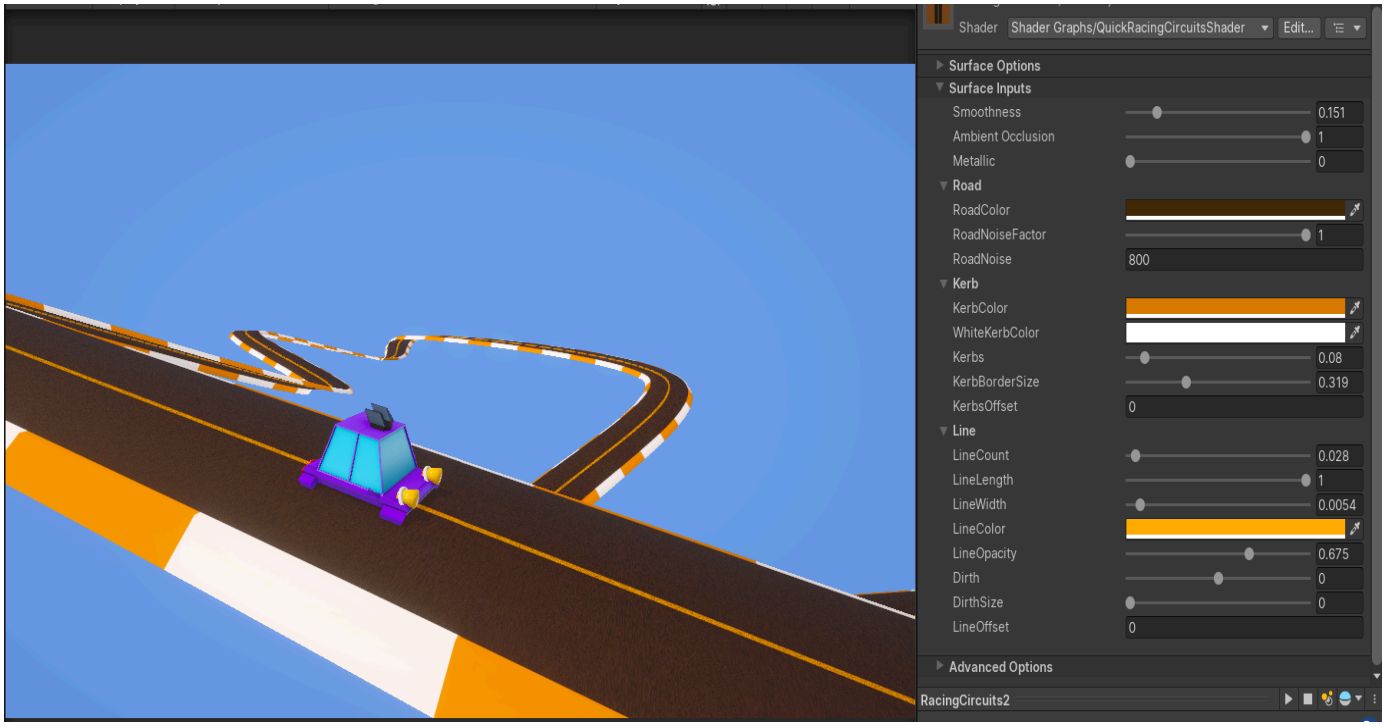

Quick Racing Circuits Material / Shader

Shader Asset Documentation: Procedural Racing Circuit Surface

Overview

This shader controls the visual properties of a road surface, including standard PBR inputs, base Road coloration, racing-style kerbs (curbs), and lane markings. It allows for real-time adjustment of road weathering and layout.

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1. Surface Inputs

Standard Physically Based Rendering (PBR) settings defining how light interacts with the road surface.

- **Smoothness (Float):** Controls the microscopic roughness of the surface.
  - *Current Value: 0.151*
  - *Usage:* Higher values make the road appear wet or polished. Lower values make the asphalt appear dry, rough, and dusty.
- **Ambient Occlusion (Float):** Controls the intensity of self-shadowing in crevices.
  - *Current Value: 1*
  - *Usage:* Set to 1 for maximum depth definition; set lower to flatten the lighting.
- **Metallic (Float):** Controls the metallic reflectivity of the surface.
  - *Current Value: 0*
  - *Usage:* Should generally remain at 0 for asphalt/concrete (dielectric materials).

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2. Road Settings

Configuration for the main asphalt surface.

- **RoadColor (Color):** The base albedo color of the asphalt.
  - *Current Value:* Dark Grey / Black
- **RoadNoiseFactor (Slider):** Controls the visibility or intensity of the grain/noise texture on the road.
  - *Current Value: 1*
- **RoadNoise (Float):** likely controls the scale or frequency of the noise pattern.
  - *Current Value: 800*
  - *Usage:* Adjusting this changes how "tight" the grain of the asphalt looks.

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3. Kerb Settings

Settings for the alternating colored edges often found on race tracks.

- **KerbColor (Color):** The primary color of the kerb (usually Red).
- **WhiteKerbColor (Color):** The secondary color of the kerb (usually White).
- **Kerbs (Slider):** Controls the repetition or frequency of the stripe pattern.
  - *Current Value: 0.7*
- **KerbBorderSize (Slider):** Defines the width or thickness of the kerb extending from the edge of the road.
  - *Current Value: 0.319*
- **KerbsOffset (Float):** Shifts the starting position of the kerb pattern along the road mesh.
  - *Current Value: 0*

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4. Line Settings

Configuration for road markings (dashed or solid lines).

- **LineCount (Slider):** Controls the density or frequency of the dashed lines.
  - *Current Value: 0.5*
- **LineLength (Slider):** Controls the length of individual dashes.
  - *Current Value: 0.656*
- **LineWidth (Slider):** Controls the thickness of the line.
  - *Current Value: 0.0087*
- **LineColor (Color):** The color of the markings.
  - *Current Value:* White
- **LineOpacity (Slider):** Controls the transparency of the paint.
  - *Current Value: 1 (Fully Opaque)*
- **Dirth (Slider):** *Likely a typo for "Dirt".* Controls the amount of grunge/dirt overlaying the lines to make them look weathered.
  - *Current Value: -0.325*
- **DirthSize (Slider):** *Likely "DirtSize".* Controls the scale of the dirt noise texture.
  - *Current Value: 2154*
- **LineOffset (Float):** Shifts the position of the line longitudinally
  - *Current Value: 0*